Rajesh Mohan Kumar

Test Architect & Senior Agentic Test Engineer

Test Architect & Senior Agentic Test Engineer with 13+ years of experience in automation (Selenium, C#, SpecFlow, API), Oracle Fusion SCM, and enterprise-grade test frameworks. Proven leader in architecting agentic QA systems, LLM-assisted test generation, and evaluation-driven GenAI workflows with a strong focus on determinism, reliability, safety, and governance. Demonstrated success in building reusable, scalable testing accelerators across Banking, Insurance, Healthcare, CRM, and Supply Chain domains.

SKILLS

Selenium WebDriver | Core Java | C# | Python | REST Assured | Agile Methodology | API | RestSharp | Cucumber | GenAI | Langchain | Jenkins | Langgraph | Maven | Postman | TestNG | Light BDD | RAG | Azure DevOps | AWS Lambda | GitHub | Langflow | DeepEval | MCP (Model Context Protocol) | Agentic QA System | LLM Evaluation

WORK EXPERIENCE

Test Architect & Senior Agentic Test Engineer

Testleaf Software Solutions Private Limited

11/2019 - Present

Responsibilities

- Architected and delivered enterprise-scale GenAI & agentic QA solutions , embedding LLMs into test design, automation, validation, and reporting workflows with production-grade reliability and governance.
- Designed and shipped RAG-based knowledge systems (pgvector, Chroma, FAISS) with golden-set evaluations, abstain policies, and retrieval recall checks to significantly reduce hallucinations and improve answer accuracy.
- Built AI-assisted test authoring pipelines using prompt templates, schema validation, and DTO/code generation to automatically produce test scenarios, data, and traceability aligned with acceptance criteria.
- Developed an AI Reporter Engine to analyze logs and test executions, detect flaky test patterns, and surface quality risk using both qualitative insights and quantitative signals.
- Led evaluation-first AI adoption by implementing continuous LLM regression, curated datasets, safety/jailbreak suites, and quality gates to maintain consistency across model updates.
- Implemented cost, safety, and privacy guardrails including token budgets, small-model routing, caching/batching, PII redaction, secure logging, and live spend dashboards.
- Drove cross-functional delivery and enablement , partnering with Dev, QA, and Platform teams to move GenAI PoCs to production with defined SLOs, rollback criteria, standardized automation frameworks, and team mentorship.
- Enabled continuous LLM regression testing with curated datasets, retrieval recall checks, automated API validations, and detailed failure analysis integrated into Azure DevOps pipelines to maintain model quality across updates.

WORK EXPERIENCE

Lead Test Engineer
Plintron Mobility Solutions

09/2011 - 10/2019

Responsibilites

- Executed comprehensive manual test cases to ensure software quality. Conducted functional, regression, and exploratory testing to identify defects.
- Developed and maintained automated test scripts using Selenium WebDriver. Enhanced test coverage and efficiency through automated testing frameworks.
- Utilized Core Java and OOPs principles to develop robust automation frameworks. Wrote clean, maintainable, and efficient code for test automation scripts.
- Implemented BDD (Behavior Driven Development) using Cucumber. Created and managed feature files, step definitions, and scenarios for automated testing.
- Used TestNG for test execution and management. Configured and executed test suites, managed test dependencies, and generated detailed test reports.
- Managed project dependencies and build processes using Maven. Configured Maven for automated build and test execution.
- Conducted API testing using Postman for manual testing of RESTful services. Automated API tests using REST Assured for validating endpoints and responses.

AI POWERED PROJECTS

User Stories to Test Case Generation (08/2024 - Present)

- **Client - Healthcare**
- Integrated RAG-based context retrieval (epics, historical defects, test standards) to enrich user story understanding and improve edge-case and negative scenario coverage.
- Implemented schema validation and structured outputs (JSON/YAML) to ensure generated test cases were automation-ready and easily consumable by Selenium, API, and BDD frameworks.
- Built traceability mapping between user stories, acceptance criteria, and generated test cases, enabling auditability and improving compliance with QA governance standards.
- Added golden-set evaluation and scoring metrics (coverage, relevance, redundancy) to continuously measure generation quality and prevent regression across prompt or model updates.
- Enabled human-in-the-loop review workflows , allowing QA leads to approve, refine, or regenerate test cases with minimal manual intervention.
- Optimized LLM cost and latency using prompt caching, token budgeting, and small-model routing for extraction steps while reserving larger models for reasoning-heavy tasks.
- Integrated with JIRA and test management tools to automatically publish approved test cases and keep QA artifacts synchronized with evolving requirements.

No-Code Agentic Test Automation using Playwright MCP (09/2022 - 06/2023)

- **Client - Banking**
- Designed intent-to-action translation layers to convert natural-language test steps into validated Playwright commands, ensuring repeatable and deterministic execution.
- Implemented tool whitelisting and permissioned MCP execution , enforcing strict control over browser, network, and data access to prevent unsafe or unintended actions.
- Built fallback and recovery strategies (retry policies, alternate locators, state revalidation) to improve resilience across dynamic UI changes.
- Integrated evaluation and confidence scoring for agent decisions, enabling early detection of ambiguous instructions and reducing execution failures.
- Enabled hybrid test authoring by allowing seamless handoff between no-code flows and code-based automation for advanced scenarios.
- Added execution telemetry and structured logs capturing agent reasoning, tool calls, and DOM interactions to accelerate RCA and compliance audits.
- Optimized cost and performance through selective model routing, prompt caching, and batched reasoning steps during large regression runs.
- Standardized reusable agent templates and guardrail policies , accelerating onboarding for QA teams and ensuring consistency across projects.

Swagger-to-API Testcase Generation Agent (03/2021 - 08/2022)

- **Client - Engineering & Construction**
- Implemented schema-aware parsing of OpenAPI components (paths, parameters, request/response models, constraints) to ensure accurate and specification-compliant test generation.
- Built RAG-based enrichment using API standards, historical defects, and contract rules to improve negative and boundary scenario quality.
- Generated data-driven and parameterized test cases , enabling scalable execution across environments and datasets.
- Integrated contract validation and response assertions (status codes, headers, schema validation, error payloads) to strengthen API reliability and backward compatibility.
- Applied evaluation-first quality gates to detect hallucinated endpoints, invalid parameters, or unsupported scenarios before test execution.
- Enabled traceability mapping between OpenAPI operations, generated test cases, and automation artifacts, improving auditability and maintenance.
- Integrated with CI/CD pipelines to auto-generate and execute API tests on spec changes, enabling early detection of breaking API changes.
- Optimized generation cost and latency using prompt caching and lightweight models for parsing, reserving larger models for reasoning-heavy edge cases.

TESTING PROJECTS

Retailer Domain
Client - Service

11/2019 - 02/2021

Responsibilities

- Designed and implemented Behavior-Driven Development (BDD) test scenarios using SpecFlow to validate application functionality against user requirements.
- Created and executed automated test scripts for RESTful APIs using RestSharp, ensuring the accuracy and reliability of API responses.
- Integrated SpecFlow and RestSharp with the existing testing framework to enhance automation capabilities and streamline test execution.
- Designed and maintained test data for SpecFlow scenarios, ensuring tests were reusable and consistent across environments.
- Worked closely with developers to identify testable requirements, resolve defects, and improve software quality through continuous feedback.
- Integrated test scripts and frameworks into Continuous Integration/Continuous Deployment (CI/CD) pipelines to enable automated testing during build and release processes.

CPOS
Client - Telecom

09/2011 - 10/2019

Responsibilities

- Thoroughly analyzed client requirements to gain a deep understanding of application functionalities.
- Developed detailed test plans and test cases for manual testing, ensuring comprehensive coverage of all application features.
- Executed manual test cases meticulously, identifying and documenting defects with precision and accuracy.
- Performed exploratory testing to identify edge cases and usability issues, enhancing user experience.
- Conducted regression testing to validate bug fixes and ensure the stability of applications. Executed automated test scripts to validate application functionalities, ensuring comprehensive test coverage.
- Provided timely and insightful feedback to stakeholders, facilitating informed decision-making and continuous improvement.

EDUCATION

B.Tech - Information Technology
Anna University

2010